SIMATSENGINEERING(SAVEETHASCHOOLOFENGINEERING)

DepartmentofComputerScienceandEngineering

ASSESSMENTTOOL3–MCQTest

CourseCode:	CSA0309	CourseTitle:	DataStructures
COAssessed:	CO3–Precision(BL3,BL5)	Assessment:	AssessmentTool3–MCQTest
Weightage:	30%ofCIA	Total Marks:	100
CO3Statement:	SolveproblemsforoptimizeddatastorageandretrievalusingBinarySearchTrees,AVLTrees,B-Trees, Red-Black Trees, and Splay Trees.		
StudentName:	Sankar R	Reg. No.:	192525131
Section/Batch:	SLOT-B	Date:	10/08/2026

CodeofConduct

I am sankar R (192525131) (Name/RegNo) certify that thissubmission ismy original work and that I have adhered to the guidelinesspecified for thisassessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: sankar R

Instructions

- Thistestcontains25MCQTestquestions(4markseach),Total:100Marks.
- W hen astudent answer using MCQ, in evaluation theanswer should bewith following requirements:

  Identifythemostappropriateanswer,conceptualunderstanding,problem-solvingability,applicationof knowledge.
- Nonegativemarking.Unansweredquestionscarryzeromarks.

Section/Topic	Focus	QNos	Marks
Preliminaries–TreeTerminology	Root,height,level,siblingsandsubtree	1–2	8
BinaryTree, BinarySearchTree	Properties,binarytreeandbinarysearchtree,insertion and deletion	3–9	28
Tree Traversals	Traversaltypes,Inorder,PostorderandPreorder	10–14	20
AVLTree	Balancingfactor,Rotation,Typesofrotation,examples	15–16	8
ApplicationsofSearchTrees	Usageofbinarytree	17	4
BTree -2-3tree,2-3-4tree	Properties,ExampleandOperations	18–19	8
TRIE	Properties,structuresandOperations	20	4
RedBlackTree	Properties,BalancingandOperations	21–22	8
SplayTree	Splaying,Typesandexamplesforinsertionanddeletion	23–25	12
	GRANDTOTAL:	100Marks	

RubricsforCalculation

Criteria	Weightage	Level4–Exemplary	Level 3 Proficient	Level 2 Developing	Level 1 Beginning
ConceptualUnderstanding	20	Demonstratesthoroughunderstanding of all coreDataStructures concepts;answersaccuratelyreflectdeepknowledge	Showsgoodunderstandinggwithminorconceptualgaps	Basicunderstanding;severalmisconceptionspresent	Limitedunderstanding; majorconceptsmisunderstood
ApplicationSkills	20	Correctlyappliesconceptstosolvecomplexandscenario-basedMCQs	Appliesconceptscorrectlyinmostmoderate-levelquestions	Struggleswith applicationincomplexscenarios	Unable to applyconceptseffectively
AnalyticalThinking	30	Consistentlyselectscorrectfhih-canswersorgorderquestions	Performswellinmoderateanalyticalquestions	Limited success inanalyticalquestions	Unabletointerptltil/reanaycaproblem-basedquestions
TimeManagement	20	Completestestefcientlywith time for review	Completestestwithintime	Strugglestocompletetestontime	Unabletocompletetest
AccuracyofResponses	10	90–100%correctanswers	75–89%correctanswers	50–74%correctanswers	Below50%correctanswers

FacultyIn-charge	CourseCoordinator	ProgramDirector
Signature:	Signature:	Signature:
Date:	Date:	Date:

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QUESTIONS

Q1
Tree Terminology
4 marks

Q2
Quad Tree
4 marks

Q3
Splay Tree
4 marks

Q4
Splay Trees
4 marks

Q5
Splay Tree
4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q8
Red-Black Tree

Q1 **Tree Terminology** 4 marks

Nodes having the same parent are called:

A. Ancestors

B. Descendants

C. Siblings

D. Leaves

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4 marks

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Splay Tree
4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q8
Red-Black Tree

Q2 **Quad Tree**

4 marks

In a Quad Tree, each internal node has at most:

A. 2 children

B. 3 children

C. 4 children

D. 8 children

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4 marks

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Splay Tree
4 marks

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Splay Trees
4 marks

Q5
Splay Tree
4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q8
Red-Black Tree

Q3 **Splay Tree**

4 marks

After accessing a node in a Splay Tree, that node is moved to:

A. Leaf

B. Middle

C. Root

D. NULL

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4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q8
Red-Black Tree

Q4 **Splay Trees** 4 marks

Identify the rotation:
50
\
70
\
90

A. Zag-Zag

B. Zig-Zig

C. Zig

D. AVL Rotation

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4 marks

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Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q8
Red-Black Tree

Q5 **Splay Tree** 4 marks

Splay Tree is a type of:

A. Complete Binary Tree

B. Self-adjusting Binary Search Tree

C. Heap Tree

D. Graph

Save Answer Next

QUESTIONS

Q1 Tree Terminology 4 marks
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Q3 Splay Tree 4 marks
Q4 Splay Trees 4 marks
Q5 Splay Tree 4 marks
Q6 Red-Black Tree 4 marks
Q7 Red-Black Tree 4 marks
Q8 Red-Black Tree 4 marks
Q9 Red-Black Tree

Q6 **Red-Black Tree** 4 marks

If a red node has a red child after insertion, what is this called?

A. Black-height problem
B. BST violation
C. Root violation
D. Red-Red violation

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QUESTIONS

Q1 Tree Terminology 4 marks
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Q3 Splay Tree 4 marks
Q4 Splay Trees 4 marks
Q5 Splay Tree 4 marks
Q6 Red-Black Tree 4 marks
Q7 Red-Black Tree 4 marks
Q8 Red-Black Tree 4 marks
Q9 Red-Black Tree

Q7 **Red-Black Tree** 4 marks

Insert sequence 70, 50, 90, 40 causes insertion of 40 under red parent. Uncle is red. What is needed?

A. Rotation
B. Recoloring
C. Deletion
D. No change

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Q1 Tree Terminology 4 marks

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Q4 Splay Trees 4 marks

Q5 Splay Tree 4 marks

Q6 Red-Black Tree 4 marks

Q7 Red-Black Tree 4 marks

Q8 Red-Black Tree 4 marks

Q9

Q8 **Red-Black Tree** 4 marks

If black height differs in two paths from the same node, the tree becomes:

A. Valid

B. Invalid

C. Balanced

D. Complete

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Q5 Splay Tree 4 marks

Q6 Red-Black Tree 4 marks

Q7 Red-Black Tree 4 marks

Q8 Red-Black Tree

Q9 **Red-Black Tree** 4 marks

Every path from a node to its descendant NULL nodes must have the same number of

A. Black nodes

B. Red nodes

C. Leaf nodes

D. Parent nodes

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4 marks

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Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q8
Red-Black Tree

Q10 **2-3 Tree** 4 marks

In a balanced 2-3 Tree, all leaves must be at:

A. Different heights

B. Root level

C. Random height

D. Same height

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Splay Tree
4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q8
Red-Black Tree

Q12 **2-3 Tree** 4 marks

The maximum number of children in a 2-3 Tree node is:

A. 3

B. 2

C. 5

D. 4

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Q5 Splay Tree 4 marks

Q6 Red-Black Tree 4 marks

Q7 Red-Black Tree 4 marks

Q8 Red-Black Tree

Q13 **TRIE** 4 marks

In Trie, 'm' represents:

A. Number of nodes

B. Length of the key

C. Number of edges

D. Height of tree

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Q4 Splay Trees 4 marks

Q5 Splay Tree 4 marks

Q6 Red-Black Tree 4 marks

Q7 Red-Black Tree 4 marks

Q14 **B-Tree** 4 marks

In a B-Tree of order 5, the maximum number of keys in a node is:

A. 4

B. 5

C. 6

D. 3

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Q5
Splay Tree
4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q15 **B-Tree** 4 marks

What is the minimum number of children for a non-root node in a B-Tree of order 'm'?

A. m/2

B. m-1

C. m

D. ⌈m/2⌉

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4 marks

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Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q8
Red-Black Tree

Q16 **Search Tree** 4 marks

The main goal of using search trees is:

A. Increase complexity

B. Reduce storage to zero

C. Improve efficiency of data operations

D. Avoid using memory

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4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q8
Red-Black Tree

Q23 **Tree Traversal**

4 marks

You are debugging a program that reconstructs the original root of a binary decision tree using traversal logs: Inorder and postorder. Given: Inorder: D B E A F C Postorder: D E B F C A What is the root?

A. A

B. B

C. C

D. D

Save Answer

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